HISTOGRAM EQUALIZATION USING OPENCV

AIM

To implement histogram equalization on a given grayscale image using OpenCV and compare the enhanced image with the original image.

PROCEDURE

1. Import the OpenCV library.
2. Read the input image using cv2.imread() in grayscale mode.
3. Check whether the image has been loaded successfully.
4. Apply histogram equalization using cv2.equalizeHist().
5. Display the original grayscale image and the equalized image.
6. Compare both images to observe the change in contrast.
7. Press any key to close the display windows.

PROGRAM

```
import cv2

def analyze_histogram(image):

    colors = ('b', 'g', 'r')

    for i, color in enumerate(colors):

        hist = cv2.calcHist([image], [i], None, [256], [0, 256])

        print(color.upper(), "histogram values:", hist[:5].flatten())

img = cv2.imread("sample.jpg")

analyze_histogram(img)
```

SAMPLE INPUT



SAMPLE OUTPUT



RESULT

Thus, histogram equalization was implemented successfully using OpenCV, and the original grayscale image was compared with the histogram-equalized image.